

Convex Quaternion Optimization for Signal Processing: Theory and Applications

Shuning Sun , Qiankun Diao , Dongpo Xu , Pauline Bourigault , and Danilo P. Mandic , *Fellow, IEEE*

Abstract—Convex optimization methods have been extensively used in communications and signal processing. However, the theory of quaternion optimization is currently not as fully developed and systematic as that of complex and real optimization. To this end, we establish the convex optimization theory in quaternion variables based on the generalized Hamilton-real (GHR) calculus. This is achieved in a way that conforms with traditional complex and real optimization theory. We present several discriminant theorems for convex quaternion functions analogous to their complex counterparts. We also provide several discriminant criteria for strongly convex functions by the theorems of convex quaternion functions. Furthermore, we prove that the quaternion Newton method can converge in one step for positive definite quadratic quaternion functions and provide two applications in quaternion signal processing. These results provide a solid theoretical foundation for convex quaternion optimization and open avenues for further developments in quaternion signal processing applications.

Index Terms—Convex quaternion functions, strongly convex quaternion functions, discriminant criterion, quaternion signal processing, GHR calculus.

I. INTRODUCTION

QUATERNIONS were first introduced by William Hamilton in 1843 as an associative but non-commutative algebra over the real numbers [1]. Over the years, quaternions have become a powerful tool in various fields, including image processing [2], [3], signal processing [4], [5], [6], and machine learning [7], [8], [9]. Examples include the work by Jia *et al.* [3], who introduced a robust method for quaternion matrix completion that can be used to reconstruct large-scale color images. Flamant *et al.* [10] demonstrated the efficiency of the Quaternion Fourier Transform in processing bivariate

signals. Ogunfunmi *et al.* [11] presented a kernel adaptive filter for quaternion data. Moreover, Mengüç *et al.* [12] designed quaternion-valued second-order Volterra adaptive filters for nonlinear 3-D and 4-D signal processing. Xia *et al.* [13] established an estimation framework for processing quaternion-valued Gaussian data. Finally, Zhang *et al.* [7] discussed a new method for reducing the computation cost of quaternion signal estimation, while Enshaieifar *et al.* [14] introduced the quaternion-valued singular spectrum analysis for multichannel electroencephalogram analysis.

The theory of real-valued and complex-valued convex optimization is well-established. It has been widely used in the areas of communications [15], machine learning [16], [17], [18] and signal processing [19], [20], [21]. In recent years, convex quaternion optimization has also attracted interest. For example, Qi *et al.* [2] studied first-order and second-order derivatives of real-valued functions of quaternion variables over their real and imaginary i , j , and k parts. However, this complicates the proof and computational process in quaternion optimization. Flamant *et al.* [22] and Liu *et al.* [23] provided the first-order characterization of convex quaternion functions by the GHR calculus [24]:

$$f(q) \geq f(p) + 4\text{Re} \left\{ \nabla_{p^*} f(p)^H (q - p) \right\}, \quad (1)$$

where $f(q) : \mathcal{C} \rightarrow \mathbb{R}$, $\mathcal{C} \subset \mathbb{H}^n$ is convex, and $\nabla_{p^*} f(p) \triangleq \left(\frac{\partial f}{\partial p^*} \right)^T$ is defined in [25]. This motivates us to strengthen further the discriminant criteria of convex quaternion functions and strongly convex quaternion functions, especially their second-order characterization.

The main challenge in extending real optimization theory to the quaternion field involves two aspects: *i*) quaternion multiplication is non-commutative; *ii*) many theories of convex quaternion optimization rely on quaternion gradients and Hessian matrices, which have only recently been defined within the framework of the GHR calculus [24], [25], [26]. The GHR calculus is a natural extension of Wirtinger calculus from the complex domain to the quaternion field [27], [28]. Before the introduction of the GHR calculus, the quaternion pseudo-derivative was used for calculating the gradient, which transforms the quaternion optimization problem into a lengthy and complicated real optimization problem; the solution is then found by using real-valued optimization algorithms [29], [30]. Flamant *et al.* [5], [22] demonstrated that the GHR calculus is a powerful theory in quaternion signal processing and

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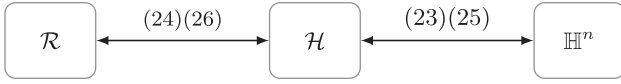


Fig. 1. Derivation process of quaternion optimization theory. (23) – (26) give the expressions of the gradient and Hessian matrix in $\mathcal{R}$, $\mathcal{H}$, and $\mathbb{H}^n$.

non-negative matrix factorization. Mengüç *et al.* [12], [31] established that the GHR calculus paves the way for the theory and applications of quaternion-valued adaptive filters. Took and Xia [32] proposed a multichannel quaternion least-mean-square based on the GHR calculus for adaptive filtering. Parcollet *et al.* [33] further emphasized the significance of the GHR calculus as a recent breakthrough in the field.

The convex quaternion optimization theory has gradually attracted attention due to its promising applications in signal processing and optimization. Our work aims to develop the convexity theory of quaternion functions using the GHR calculus [24]. To this end, we make use of the duality of the augmented quaternion vector $\mathbf{q}_{\mathcal{H}} \triangleq (\mathbf{q}^T, \mathbf{q}^{iT}, \mathbf{q}^{jT}, \mathbf{q}^{kT})^T$ (see (3)) and the augmented real vector $\mathbf{q}_{\mathcal{R}} \triangleq (\mathbf{q}_a^T, \mathbf{q}_b^T, \mathbf{q}_c^T, \mathbf{q}_d^T)^T$, where the quaternion vector $\mathbf{q} = \mathbf{q}_a + \mathbf{q}_b i + \mathbf{q}_c j + \mathbf{q}_d k \in \mathbb{H}^n$ [25]. Next, we employ the relationships between augmented quaternion gradient and augmented real gradient, as well as between the augmented quaternion Hessian matrix and augmented real Hessian matrix, as shown in [25]. This allows us to establish the discriminant criteria for convexity, from the real field $\mathcal{R}$ to the augmented quaternion space $\mathcal{H}$, and then to the quaternion field $\mathbb{H}^n$, where the sets $\mathcal{R}$ and $\mathcal{H}$ are defined in (5) and (6), as illustrated in Fig. 1. Furthermore, we develop several discriminant theorems for strong convexity based on the discriminant criteria of convex quaternion functions. Finally, we prove the quadratic termination of the quaternion Newton method for positive definite quadratic quaternion functions and provide two illustrative applications in the field of quaternion signal processing, including quaternion collaborative representation-based classification (QCRC) and quaternion binary logistic regression (QBLR). Numerical experiments also support our theoretical results.

This work establishes the theory of convex optimization in the quaternion field, whereby:

- By using the GHR calculus, we establish several discriminant theorems for convex functions in the quaternion field. These theorems include gradient monotonicity and second-order characterization.
- We provide several discriminant criteria for strongly convex quaternion functions based on the theorems of convex functions. These are consistent with the real and complex correspondence criteria.
- We prove that the quaternion Newton method can converge in one step for positive definite quadratic functions. Additionally, we provide two applications of convex quaternion optimization to support theoretical analysis.

This paper is organized as follows. Section II gives an overview of quaternion algebra and the GHR calculus. Section III presents several discriminant theorems for convex quaternion functions, covering gradient monotonicity, second-order characterization, and an example of a convex quaternion

function. Section IV introduces the discriminant theorems for strongly convex quaternion functions and the proof of quadratic termination. In Section V, we provide two practical applications in quaternion signal processing. Finally, this paper concludes with Section VI.

II. PRELIMINARIES

A. Quaternion Algebra

A quaternion q , can be expressed as

$$q = q_a + q_b i + q_c j + q_d k, \quad (2)$$

where $q_a, q_b, q_c, q_d \in \mathbb{R}$, and the imaginary units i, j and k satisfy $i^2 = j^2 = k^2 = ijk = -1$, $ij = -ji = k$, $jk = -kj = i$, $ki = -ik = j$. The set of quaternions is defined as $\mathbb{H} \triangleq \{q = q_a + q_b i + q_c j + q_d k \mid q_a, q_b, q_c, q_d \in \mathbb{R}\}$. Owing to the properties of the imaginary units, the multiplication of two quaternions in $\mathbb{H}$ is noncommutative. The real part of q is denoted by $\text{Re}\{q\} = q_a$, whereas the imaginary part is $\text{Im}\{q\} = q_b i + q_c j + q_d k$. The conjugate of q is $q^* = \text{Re}\{q\} - \text{Im}\{q\} = q_a - q_b i - q_c j - q_d k$. The modulus of q is defined as $|q| = \sqrt{qq^*}$. If $q \neq 0$, the inverse of q is $q^{-1} = \frac{q^*}{|q|^2}$. The rotation of a quaternion q is defined as [34]

$$q^\mu \triangleq \mu q \mu^{-1} \quad (3)$$

where μ is a nonzero quaternion. For any $p, q \in \mathbb{H}$, and $\forall \nu, \mu \in \mathbb{H}$, the following holds [26]

$$\begin{aligned} c(pq)^\mu &= p^\mu q^\mu, \quad q^{\mu\nu} = (q^\nu)^\mu, \\ q^{\mu*} &\triangleq (q^*)^\mu = (q^\mu)^* \triangleq q^{*\mu}. \end{aligned} \quad (4)$$

Consider a quaternion vector $\mathbf{q} = \mathbf{q}_a + \mathbf{q}_b i + \mathbf{q}_c j + \mathbf{q}_d k \in \mathbb{H}^n$ where $\mathbf{q}_a, \mathbf{q}_b, \mathbf{q}_c, \mathbf{q}_d \in \mathbb{R}^n$. Define its augmented real vector as $\mathbf{q}_{\mathcal{R}} \triangleq (\mathbf{q}_a^T, \mathbf{q}_b^T, \mathbf{q}_c^T, \mathbf{q}_d^T)^T \in \mathcal{R}$ [4], [35] and the augmented quaternion vector as $\mathbf{q}_{\mathcal{H}} \triangleq (\mathbf{q}^T, \mathbf{q}^{iT}, \mathbf{q}^{jT}, \mathbf{q}^{kT})^T \in \mathcal{H}$ [25], where the set of augmented real vectors and the set of augmented quaternion vectors are defined as

$$\mathcal{R} \triangleq \{\mathbf{q}_{\mathcal{R}} = (\mathbf{q}_a^T, \mathbf{q}_b^T, \mathbf{q}_c^T, \mathbf{q}_d^T)^T \mid \mathbf{q} \in \mathbb{H}^n\} = \mathbb{R}^{4n}, \quad (5)$$

$$\mathcal{H} \triangleq \{\mathbf{q}_{\mathcal{H}} = (\mathbf{q}^T, \mathbf{q}^{iT}, \mathbf{q}^{jT}, \mathbf{q}^{kT})^T \mid \mathbf{q} \in \mathbb{H}^n\} \subset \mathbb{H}^{4n}. \quad (6)$$

By definition, there exists a one-to-one mapping between $\mathbb{H}^n$, $\mathcal{R}$ and $\mathcal{H}$ [22]. The 2-norms of $\mathbf{q}$, $\mathbf{q}_{\mathcal{R}}$ and $\mathbf{q}_{\mathcal{H}}$ are defined as $\|\mathbf{q}\|_2 \triangleq \sqrt{\mathbf{q}^H \mathbf{q}}$, $\|\mathbf{q}_{\mathcal{R}}\|_2 \triangleq \sqrt{\mathbf{q}_{\mathcal{R}}^T \mathbf{q}_{\mathcal{R}}}$, and $\|\mathbf{q}_{\mathcal{H}}\|_2 \triangleq \sqrt{\mathbf{q}_{\mathcal{H}}^H \mathbf{q}_{\mathcal{H}}}$.

The relationship between the augmented quaternion vector $\mathbf{q}_{\mathcal{H}}$ and the augmented real vector $\mathbf{q}_{\mathcal{R}}$ is given by [25]

$$\mathbf{q}_{\mathcal{H}} = \mathbf{J}_n \mathbf{q}_{\mathcal{R}} \quad \Leftrightarrow \quad \mathbf{q}_{\mathcal{R}} = \frac{1}{4} \mathbf{J}_n^H \mathbf{q}_{\mathcal{H}}, \quad (7)$$

where

$$\mathbf{J}_n = \begin{pmatrix} \mathbf{I}_n & i\mathbf{I}_n & j\mathbf{I}_n & k\mathbf{I}_n \\ \mathbf{I}_n & i\mathbf{I}_n & -j\mathbf{I}_n & -k\mathbf{I}_n \\ \mathbf{I}_n & -i\mathbf{I}_n & j\mathbf{I}_n & -k\mathbf{I}_n \\ \mathbf{I}_n & -i\mathbf{I}_n & -j\mathbf{I}_n & k\mathbf{I}_n \end{pmatrix} \in \mathbb{H}^{4n \times 4n}, \quad (8)$$

and $\mathbf{J}_n^H \mathbf{J}_n = 4\mathbf{I}_{4n}$, $\mathbf{I}_n$ is the $n \times n$ identity matrix, and $\mathbf{J}_n^H$ is the conjugate transpose of $\mathbf{J}_n$.

Proposition 2.1: For any $\mathbf{p}, \mathbf{q} \in \mathbb{H}^n$, their augmented real vectors are $\mathbf{p}_{\mathcal{R}}, \mathbf{q}_{\mathcal{R}} \in \mathcal{R}$, and their augmented quaternion vectors are $\mathbf{p}_{\mathcal{H}}, \mathbf{q}_{\mathcal{H}} \in \mathcal{H}$. Then

$$(a) \quad \mathbf{p}_{\mathcal{H}}^{\top} \mathbf{q}_{\mathcal{H}} = 4\text{Re}\{\mathbf{p}^{\top} \mathbf{q}\}; \quad (9)$$

$$(b) \quad 4\mathbf{p}_{\mathcal{R}}^{\top} \mathbf{q}_{\mathcal{R}} = \mathbf{p}_{\mathcal{H}}^{\text{H}} \mathbf{q}_{\mathcal{H}} = 4\text{Re}\{\mathbf{p}^{\text{H}} \mathbf{q}\}; \quad (10)$$

$$(c) \quad 2\|\mathbf{p}_{\mathcal{R}}\|_2 = \|\mathbf{p}_{\mathcal{H}}\|_2 = 2\|\mathbf{p}\|_2; \quad (11)$$

$$(d) \quad \|\mathbf{p} + \mathbf{q}\|_2^2 = \|\mathbf{p}\|_2^2 + 2\text{Re}\{\mathbf{p}^{\text{H}} \mathbf{q}\} + \|\mathbf{q}\|_2^2. \quad (12)$$

Proof: By the relationship of $\mathbf{q}$, $\mathbf{q}_{\mathcal{R}}$, and $\mathbf{q}_{\mathcal{H}}$, we have

$$(a) \quad \mathbf{p}_{\mathcal{H}}^{\top} \mathbf{q}_{\mathcal{H}} = \sum_{\mu \in \{1, i, j, k\}} \mathbf{p}^{\mu\top} \mathbf{q}^{\mu} \stackrel{(4)}{=} \sum_{\mu \in \{1, i, j, k\}} (\mathbf{p}^{\top} \mathbf{q})^{\mu} \stackrel{(7)}{=} 4\text{Re}\{\mathbf{p}^{\top} \mathbf{q}\}; \quad (13)$$

$$(b) \quad 4\mathbf{p}_{\mathcal{R}}^{\top} \mathbf{q}_{\mathcal{R}} = 4\mathbf{p}_{\mathcal{H}}^{\text{H}} \mathbf{q}_{\mathcal{H}} \stackrel{(7)}{=} \mathbf{p}_{\mathcal{H}}^{\text{H}} \mathbf{J}_n \frac{1}{4} \mathbf{J}_n^{\text{H}} \mathbf{q}_{\mathcal{H}} = \mathbf{p}_{\mathcal{H}}^{\text{H}} \mathbf{q}_{\mathcal{H}} \stackrel{(9)}{=} 4\text{Re}\{\mathbf{p}^{\text{H}} \mathbf{q}\}; \quad (14)$$

$$(c) \quad 4\|\mathbf{p}_{\mathcal{R}}\|_2^2 = 4\mathbf{p}_{\mathcal{R}}^{\top} \mathbf{p}_{\mathcal{R}} \stackrel{(10)}{=} \mathbf{p}_{\mathcal{H}}^{\text{H}} \mathbf{p}_{\mathcal{H}} = \|\mathbf{p}_{\mathcal{H}}\|_2^2 \stackrel{(10)}{=} 4\text{Re}\{\mathbf{p}^{\text{H}} \mathbf{p}\} = 4\mathbf{p}^{\text{H}} \mathbf{p} = 4\|\mathbf{p}\|_2^2; \quad (15)$$

$$(d) \quad \|\mathbf{p} + \mathbf{q}\|_2^2 = (\mathbf{p} + \mathbf{q})^{\text{H}} (\mathbf{p} + \mathbf{q}) = \mathbf{p}^{\text{H}} \mathbf{p} + \mathbf{p}^{\text{H}} \mathbf{q} + \mathbf{q}^{\text{H}} \mathbf{p} + \mathbf{q}^{\text{H}} \mathbf{q} = \|\mathbf{p}\|_2^2 + 2\text{Re}\{\mathbf{p}^{\text{H}} \mathbf{q}\} + \|\mathbf{q}\|_2^2. \quad (16)$$

This completes the proof. $\square$

B. GHR Calculus

A quaternion function $f(\mathbf{q}) = f_a(q_a, q_b, q_c, q_d) + f_b(q_a, q_b, q_c, q_d)i + f_c(q_a, q_b, q_c, q_d)j + f_d(q_a, q_b, q_c, q_d)k$ is called real-differentiable, if f_a, f_b, f_c, f_d are differentiable as functions of the real variables q_a, q_b, q_c, q_d [36].

Definition 2.1: (GHR derivatives [24]). If $f: \mathbb{H} \rightarrow \mathbb{H}$ is real-differentiable, then the left GHR derivatives of the function f with respect to q^{μ} and $q^{\mu*}$ ($\mu \neq 0, \mu \in \mathbb{H}$) are defined as

$$\frac{\partial f}{\partial q^{\mu}} = \frac{1}{4} \left(\frac{\partial f}{\partial q_a} - \frac{\partial f}{\partial q_b} i^{\mu} - \frac{\partial f}{\partial q_c} j^{\mu} - \frac{\partial f}{\partial q_d} k^{\mu} \right) \in \mathbb{H}, \quad (17)$$

$$\frac{\partial f}{\partial q^{\mu*}} = \frac{1}{4} \left(\frac{\partial f}{\partial q_a} + \frac{\partial f}{\partial q_b} i^{\mu} + \frac{\partial f}{\partial q_c} j^{\mu} + \frac{\partial f}{\partial q_d} k^{\mu} \right) \in \mathbb{H}, \quad (18)$$

where $\mathbf{q} = q_a + q_b i + q_c j + q_d k$, $q_a, q_b, q_c, q_d \in \mathbb{R}$, and $\frac{\partial f}{\partial q_a}, \frac{\partial f}{\partial q_b}, \frac{\partial f}{\partial q_c}, \frac{\partial f}{\partial q_d} \in \mathbb{H}$ are the partial derivatives of f with respect to q_a, q_b, q_c, q_d .

If $f: \mathbb{H} \rightarrow \mathbb{R}, g: \mathbb{H} \rightarrow \mathbb{R}$, the following rules hold [24]

$$\frac{\partial(fg)}{\partial q^{\mu}} = f \frac{\partial g}{\partial q^{\mu}} + \frac{\partial f}{\partial q^{\mu}} g, \quad \frac{\partial(fg)}{\partial q^{\mu*}} = f \frac{\partial g}{\partial q^{\mu*}} + \frac{\partial f}{\partial q^{\mu*}} g; \quad (19)$$

$$\frac{\partial f(g(q))}{\partial q^{\mu}} = f'(g) \frac{\partial g}{\partial q^{\mu}}, \quad \frac{\partial f(g(q))}{\partial q^{\mu*}} = f'(g) \frac{\partial g}{\partial q^{\mu*}}; \quad (20)$$

$$\left(\frac{\partial f}{\partial q^{\mu}} \right)^{\nu} = \frac{\partial f}{\partial q^{\nu\mu}}, \quad \left(\frac{\partial f}{\partial q^{\mu*}} \right)^{\nu} = \frac{\partial f}{\partial q^{\nu\mu*}}; \quad (21)$$

$$\left(\frac{\partial f}{\partial q^{\mu}} \right)^{*} = \frac{\partial f}{\partial q^{\mu*}}, \quad \left(\frac{\partial f}{\partial q^{\mu*}} \right)^{*} = \frac{\partial f}{\partial q^{\mu}}. \quad (22)$$

For a real-differentiable function $f(\mathbf{q}): \mathbb{H}^n \rightarrow \mathbb{R}$, the augmented real gradient and the augmented quaternion gradient are respectively defined as $\nabla_{\mathcal{R}} f \triangleq \left(\frac{\partial f}{\partial \mathbf{q}_{\mathcal{R}}} \right)^{\top}$ and

$$\nabla_{\mathcal{H}} f \triangleq \left(\frac{\partial f}{\partial \mathbf{q}_{\mathcal{H}}} \right)^{\top} = \begin{pmatrix} \nabla_{\mathbf{q}} f \\ \nabla_{\mathbf{q}^i} f \\ \nabla_{\mathbf{q}^j} f \\ \nabla_{\mathbf{q}^k} f \end{pmatrix}, \quad (23)$$

where $\nabla_{\mathbf{q}} f \triangleq \left(\frac{\partial f}{\partial \mathbf{q}} \right)^{\top}$ [25]. Their relationship is given by [37]

$$\nabla_{\mathcal{H}^*} f = \frac{1}{4} \mathbf{J}_n \nabla_{\mathcal{R}} f \Leftrightarrow \nabla_{\mathcal{R}} f = \mathbf{J}_n^{\text{H}} \nabla_{\mathcal{H}^*} f, \quad (24)$$

where $\nabla_{\mathcal{H}^*} f = (\nabla_{\mathcal{H}} f)^*$. If $f(\mathbf{q})$ is also second-order continuous real-differentiable, the augmented real Hessian matrix, and the augmented quaternion Hessian matrix are respectively defined as $\mathbf{H}_{\mathcal{R}\mathcal{R}} \triangleq \frac{\partial}{\partial \mathbf{q}_{\mathcal{R}}} \left(\frac{\partial f}{\partial \mathbf{q}_{\mathcal{R}}} \right)^{\top}$, and

$$\mathbf{H}_{\mathcal{H}\mathcal{H}^*} \triangleq \frac{\partial}{\partial \mathbf{q}_{\mathcal{H}}} \left(\frac{\partial f}{\partial \mathbf{q}_{\mathcal{H}}^*} \right)^{\top} = \begin{pmatrix} \mathbf{H}_{\mathbf{q}\mathbf{q}^*} & \mathbf{H}_{\mathbf{q}^i\mathbf{q}^*} & \mathbf{H}_{\mathbf{q}^j\mathbf{q}^*} & \mathbf{H}_{\mathbf{q}^k\mathbf{q}^*} \\ \mathbf{H}_{\mathbf{q}\mathbf{q}^{i*}} & \mathbf{H}_{\mathbf{q}^i\mathbf{q}^{i*}} & \mathbf{H}_{\mathbf{q}^j\mathbf{q}^{i*}} & \mathbf{H}_{\mathbf{q}^k\mathbf{q}^{i*}} \\ \mathbf{H}_{\mathbf{q}\mathbf{q}^{j*}} & \mathbf{H}_{\mathbf{q}^i\mathbf{q}^{j*}} & \mathbf{H}_{\mathbf{q}^j\mathbf{q}^{j*}} & \mathbf{H}_{\mathbf{q}^k\mathbf{q}^{j*}} \\ \mathbf{H}_{\mathbf{q}\mathbf{q}^{k*}} & \mathbf{H}_{\mathbf{q}^i\mathbf{q}^{k*}} & \mathbf{H}_{\mathbf{q}^j\mathbf{q}^{k*}} & \mathbf{H}_{\mathbf{q}^k\mathbf{q}^{k*}} \end{pmatrix}, \quad (25)$$

where $\mathbf{H}_{\mathbf{q}\mathbf{q}^*} \triangleq \frac{\partial}{\partial \mathbf{q}} \left(\frac{\partial f}{\partial \mathbf{q}^*} \right)^{\top}$. Their relationship is that [25]

$$\mathbf{H}_{\mathcal{H}\mathcal{H}^*} = \frac{1}{16} \mathbf{J}_n \mathbf{H}_{\mathcal{R}\mathcal{R}} \mathbf{J}_n^{\text{H}} \Leftrightarrow \mathbf{H}_{\mathcal{R}\mathcal{R}} = \mathbf{J}_n^{\text{H}} \mathbf{H}_{\mathcal{H}\mathcal{H}^*} \mathbf{J}_n. \quad (26)$$

Since (26) and $\mathbf{H}_{\mathcal{R}\mathcal{R}}$ is symmetric, $\mathbf{H}_{\mathcal{H}\mathcal{H}^*}$ is Hermitian.

Remark 1: Notice that we can also define two other quaternion Hessian matrices $\mathbf{H}_{\mathbf{q}^* \mathbf{q}}$ and $\mathbf{H}_{\mathbf{q}^* \mathbf{q}^*}$, but for these it is difficult to obtain the relationship between the augmented quaternion Hessian matrix and its real counterpart (see (26)). So they are omitted.

Proposition 2.2: If the quaternion function $f(\mathbf{q}): \mathbb{H}^n \rightarrow \mathbb{R}$ is real-differentiable, then $\forall \mathbf{p}, \mathbf{q} \in \mathbb{H}^n$ we have

$$(a) \quad \nabla_{\mathcal{R}} f(\mathbf{p}_{\mathcal{R}})^{\top} \mathbf{q}_{\mathcal{R}} = \nabla_{\mathcal{H}^*} f(\mathbf{p}_{\mathcal{H}})^{\text{H}} \mathbf{q}_{\mathcal{H}} = 4\text{Re}\{\nabla_{\mathbf{p}^*} f(\mathbf{p})^{\text{H}} \mathbf{q}\}; \quad (27)$$

$$(b) \quad \nabla_{\mathcal{R}} f(\mathbf{p}_{\mathcal{R}})^{\top} \nabla_{\mathcal{R}} f(\mathbf{q}_{\mathcal{R}}) = 4\nabla_{\mathcal{H}^*} f(\mathbf{p}_{\mathcal{H}})^{\text{H}} \nabla_{\mathcal{H}^*} f(\mathbf{q}_{\mathcal{H}}) = 16\text{Re}\{\nabla_{\mathbf{p}^*} f(\mathbf{p})^{\text{H}} \nabla_{\mathbf{q}^*} f(\mathbf{q})\}; \quad (28)$$

$$(c) \quad \|\nabla_{\mathcal{R}} f(\mathbf{p}_{\mathcal{R}})\|_2 = 2\|\nabla_{\mathcal{H}} f(\mathbf{p}_{\mathcal{H}})\|_2 = 4\|\nabla_{\mathbf{p}} f(\mathbf{p})\|_2. \quad (29)$$

Proof: By (7), (24), $\mathbf{J}_n^{\text{H}} \mathbf{J}_n = 4\mathbf{I}_{4n}$, and $\nabla_{\mathcal{R}} f \in \mathbb{R}^n$, we have

$$(a) \quad \nabla_{\mathcal{R}} f(\mathbf{p}_{\mathcal{R}})^{\top} \mathbf{q}_{\mathcal{R}} = \nabla_{\mathcal{R}} f(\mathbf{p}_{\mathcal{R}})^{\text{H}} \mathbf{q}_{\mathcal{R}} \stackrel{(7)(24)}{=} \nabla_{\mathcal{H}^*} f(\mathbf{p}_{\mathcal{H}})^{\text{H}} \mathbf{J}_n \frac{1}{4} \mathbf{J}_n^{\text{H}} \mathbf{q}_{\mathcal{H}} = \nabla_{\mathcal{H}^*} f(\mathbf{p}_{\mathcal{H}})^{\text{H}} \mathbf{q}_{\mathcal{H}} = \sum_{\mu \in \{1, i, j, k\}} \nabla_{\mathbf{p}^*} f(\mathbf{p})^{\mu\text{H}} \mathbf{q}^{\mu}$$

$$\begin{aligned}
&\stackrel{(4)}{=} \sum_{\mu \in \{1, i, j, k\}} \left(\nabla_{\mathbf{p}^*} f(\mathbf{p})^H \mathbf{q} \right)^\mu \\
&\stackrel{(7)}{=} 4\text{Re} \left\{ \nabla_{\mathbf{p}^*} f(\mathbf{p})^H \mathbf{q} \right\}; \quad (30)
\end{aligned}$$

$$\begin{aligned}
(b) \quad &\nabla_{\mathcal{R}} f(\mathbf{p}_{\mathcal{R}})^T \nabla_{\mathcal{R}} f(\mathbf{q}_{\mathcal{R}}) = \nabla_{\mathcal{R}} f(\mathbf{p}_{\mathcal{R}})^H \nabla_{\mathcal{R}} f(\mathbf{q}_{\mathcal{R}}) \\
&\stackrel{(24)}{=} \nabla_{\mathcal{H}^*} f(\mathbf{p}_{\mathcal{H}})^H \mathbf{J}_n \mathbf{J}_n^H \nabla_{\mathcal{H}^*} f(\mathbf{q}_{\mathcal{H}}) \\
&= 4 \nabla_{\mathcal{H}^*} f(\mathbf{p}_{\mathcal{H}})^H \nabla_{\mathcal{H}^*} f(\mathbf{q}_{\mathcal{H}}) \\
&= 4 \sum_{\mu \in \{1, i, j, k\}} \nabla_{\mathbf{p}^*} f(\mathbf{p})^{\mu H} \nabla_{\mathbf{q}^*} f(\mathbf{q})^\mu \\
&\stackrel{(7)}{=} 16\text{Re} \left\{ \nabla_{\mathbf{p}^*} f(\mathbf{p})^H \nabla_{\mathbf{q}^*} f(\mathbf{q}) \right\}; \quad (31)
\end{aligned}$$

(c) Let $\mathbf{q} = \mathbf{p}$ in (28). $\square$

III. DISCRIMINANT THEOREMS FOR CONVEX QUATERNION FUNCTIONS

This section aims to introduce several discriminant criteria for convex quaternion functions, including gradient monotonicity and second-order characterization. An example illustrates how these criteria are applied in practice.

A. Convex Set and Convex Function

We begin by introducing the fundamental concepts, such as those of a convex set and convex function, as in Section 2.1 of [38]. The set $\mathcal{C}$ is called convex, if $\forall \mathbf{x}, \mathbf{y} \in \mathcal{C}, \forall \theta \in [0, 1], \theta \mathbf{x} + (1 - \theta) \mathbf{y} \in \mathcal{C}$. The set $\mathcal{C}$ can be a subset of $\mathbb{H}^n, \mathcal{R}$ or $\mathcal{H}$. The sets $\mathcal{C} \subset \mathbb{H}^n, \mathcal{C}_{\mathcal{R}} \triangleq \left\{ \mathbf{q}_{\mathcal{R}} = (\mathbf{q}_a^T, \mathbf{q}_b^T, \mathbf{q}_c^T, \mathbf{q}_d^T)^T \mid \mathbf{q} \in \mathcal{C} \right\} \subset \mathcal{R} = \mathbb{R}^{4n}$, and $\mathcal{C}_{\mathcal{H}} \triangleq \left\{ \mathbf{q}_{\mathcal{H}} = (\mathbf{q}^T, \mathbf{q}^{i^T}, \mathbf{q}^{j^T}, \mathbf{q}^{k^T})^T \mid \mathbf{q} \in \mathcal{C} \right\} \subset \mathcal{H} \subset \mathbb{H}^{4n}$ are related as follows [22]

$$\mathcal{C} \text{ is convex} \Leftrightarrow \mathcal{C}_{\mathcal{R}} \text{ is convex} \Leftrightarrow \mathcal{C}_{\mathcal{H}} \text{ is convex.} \quad (32)$$

A function f is said to be convex, if the domain of $f(\text{dom} f)$ is convex, and $\forall \mathbf{x}, \mathbf{y} \in \text{dom} f, \forall \theta \in [0, 1]$,

$$f(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta) f(\mathbf{y}). \quad (33)$$

The range of function f is the real domain $\mathbb{R}$, and $\text{dom} f$ can be a subset of $\mathbb{H}^n, \mathcal{R}$ or $\mathcal{H}$.

B. First-Order Characterization of Discriminant Theorems for Convex Quaternion Functions

A popular first-order characterization was given in [22], [23] and in (1). The other is gradient monotonicity, as follows.

Theorem 3.1: (Gradient monotonicity). Consider a convex set $\mathcal{C} \subset \mathbb{H}^n$ and a real-differentiable quaternion function $f(\mathbf{q}) : \mathcal{C} \rightarrow \mathbb{R}$. Then, $f(\mathbf{q})$ is convex if and only if (shortened to iff) $\forall \mathbf{p}, \mathbf{q} \in \mathcal{C}$,

$$\text{Re} \left\{ (\nabla_{\mathbf{p}^*} f(\mathbf{p}) - \nabla_{\mathbf{q}^*} f(\mathbf{q}))^H (\mathbf{p} - \mathbf{q}) \right\} \geq 0. \quad (34)$$

Proof: From (32), $\mathcal{C}$ is convex iff $\mathcal{C}_{\mathcal{R}}$ is convex. We already know from Section 2.1 of [38] that for a differentiable real function, $f(\mathbf{q}_{\mathcal{R}})$ is convex iff $\forall \mathbf{p}_{\mathcal{R}}, \mathbf{q}_{\mathcal{R}} \in \mathcal{C}_{\mathcal{R}}$,

$$(\nabla_{\mathcal{R}} f(\mathbf{p}_{\mathcal{R}}) - \nabla_{\mathcal{R}} f(\mathbf{q}_{\mathcal{R}}))^T (\mathbf{p}_{\mathcal{R}} - \mathbf{q}_{\mathcal{R}}) \geq 0, \quad (35)$$

where the set $\mathcal{C}_{\mathcal{R}} \subset \mathcal{R}$ is convex. Therefore, we have

$$\begin{aligned}
&(\nabla_{\mathcal{R}} f(\mathbf{p}_{\mathcal{R}}) - \nabla_{\mathcal{R}} f(\mathbf{q}_{\mathcal{R}}))^T (\mathbf{p}_{\mathcal{R}} - \mathbf{q}_{\mathcal{R}}) \\
&\stackrel{(27)}{=} 4\text{Re} \left\{ (\nabla_{\mathbf{p}^*} f(\mathbf{p}) - \nabla_{\mathbf{q}^*} f(\mathbf{q}))^H (\mathbf{p} - \mathbf{q}) \right\}. \quad (36)
\end{aligned}$$

Upon substituting (36) into (35), the proof follows. $\square$

Since the following two theorems do not involve the interchange of products, quaternion gradients, or Hessian matrices, these can be easily derived using the methods from the real domain, as in Section 3.1 of [38] and Section 3.1 of [39].

Definition 3.1: (Epigraph). For the quaternion generalized real-valued function $f(\mathbf{q}) : \mathbb{H}^n \rightarrow \mathbb{R} \cup \{\pm\infty\}$, the set

$$\text{epi} f = \{(\mathbf{q}, t) \in \mathbb{H}^{n+1} \mid f(\mathbf{q}) \leq t, t \in \mathbb{R}\} \quad (37)$$

is called the epigraph of $f(\mathbf{q})$.

Theorem 3.2: The quaternion generalized real-valued function $f(\mathbf{q}) : \mathcal{C} \subset \mathbb{H}^n \rightarrow \mathbb{R} \cup \{\pm\infty\}$ is convex, iff $\text{epi} f$ is a convex set.

Theorem 3.3: The quaternion function $f(\mathbf{q}) : \mathcal{C} \subset \mathbb{H}^n \rightarrow \mathbb{R}$ is convex iff $\forall \mathbf{q} \in \mathcal{C}, \mathbf{v} \in \mathbb{H}^n, g : \mathcal{S} \rightarrow \mathbb{R}$, the function

$$g(t) = f(\mathbf{q} + t\mathbf{v}) \quad (38)$$

is convex, where $\mathcal{S} \triangleq \{t \in \mathbb{R} \mid \mathbf{q} + t\mathbf{v} \in \mathcal{C}\} \subset \mathbb{R}$.

C. Second-Order Characterization of Discriminant Theorems for Convex Quaternion Functions

Before introducing the second-order characterization of convex quaternion functions, we must define positive definite quaternion matrices.

Definition 3.2: (Positive definite matrix). The matrix $\mathbf{A} \in \mathbb{H}^{n \times n}$ is called positive definite, if

$$\text{Re} \{ \mathbf{x}^H \mathbf{A} \mathbf{x} \} > 0, \quad \forall \mathbf{x} \in \mathbb{H}^n, \mathbf{x} \neq \mathbf{0}, \quad (39)$$

and is denoted by $\mathbf{A} \succ \mathbf{O}$, where $\mathbf{O}$ is $n \times n$ zero matrix. Similarly, the matrix $\mathbf{A} \in \mathbb{H}^{n \times n}$ is called positive semi-definite, if

$$\text{Re} \{ \mathbf{x}^H \mathbf{A} \mathbf{x} \} \geq 0, \quad \forall \mathbf{x} \in \mathbb{H}^n, \mathbf{x} \neq \mathbf{0}, \quad (40)$$

and is denoted by $\mathbf{A} \succeq \mathbf{O}$.

Theorem 3.4: If the matrix $\mathbf{A} \in \mathbb{H}^{n \times n}$ satisfies $\mathbf{A}^H = \mathbf{A}$, then $\mathbf{A}$ is positive definite (positive semi-definite) iff

$$\mathbf{x}^H \mathbf{A} \mathbf{x} > 0 \text{ (} \mathbf{x}^H \mathbf{A} \mathbf{x} \geq 0 \text{)}, \quad \forall \mathbf{x} \in \mathbb{H}^n, \mathbf{x} \neq \mathbf{0}. \quad (41)$$

Proof: This is straightforward to prove by applying Definition 3.2. $\square$

If a quaternion function $f(\mathbf{q})$ is second-order continuous real-differentiable, the Hessian matrix can be used to discriminate its convexity, as shown below.

Theorem 3.5: (Second-order characterization). Consider a convex set $\mathcal{C} \subset \mathbb{H}^n$ and a second-order continuous real-differentiable quaternion function $f(\mathbf{q}) : \mathcal{C} \rightarrow \mathbb{R}$. Then, the following three statements are equivalent:

- (a) $f(\mathbf{q})$ is convex;
- (b) $\mathbf{H}_{\mathcal{H}\mathcal{H}^*} \succeq \mathbf{O}$;
- (c) $\sum_{\nu \in \{1, i, j, k\}} \text{Re} \{ \mathbf{x}^H \mathbf{H}_{\mathbf{q}^{\nu} \mathbf{q}^*} \mathbf{x}^{\nu} \} \geq 0, \forall \mathbf{x} \in \mathbb{H}^n, \mathbf{x} \neq \mathbf{0}$.

Proof: We first show that (a) is equivalent to (b). From (32), the set $\mathcal{C}$ is convex iff the set $\mathcal{C}_{\mathcal{R}}$ is convex. We already know from Section 2.1 of [38] that for a second-order continuous differentiable function, $f(q_{\mathcal{R}})$ is convex iff

$$H_{\mathcal{R}\mathcal{R}} \succeq \mathbf{O}, \quad \forall q_{\mathcal{R}} \in \mathcal{C}_{\mathcal{R}}, \quad (42)$$

where $\mathcal{C}_{\mathcal{R}} \subset \mathcal{R}$ is convex. Then, $\forall x_{\mathcal{H}} \in \mathcal{H}$, $x_{\mathcal{H}} \neq \mathbf{0}$, we have

$$\begin{aligned} x_{\mathcal{H}}^H H_{\mathcal{H}\mathcal{H}^*} x_{\mathcal{H}} &\stackrel{(26)}{=} \frac{1}{16} x_{\mathcal{H}}^H J_n H_{\mathcal{R}\mathcal{R}} J_n^H x_{\mathcal{H}} \\ &= \frac{1}{16} \left(J_n^H x_{\mathcal{H}} \right)^H H_{\mathcal{R}\mathcal{R}} \left(J_n^H x_{\mathcal{H}} \right) \\ &\stackrel{(7)}{=} x_{\mathcal{R}}^H H_{\mathcal{R}\mathcal{R}} x_{\mathcal{R}}. \end{aligned} \quad (43)$$

Since $H_{\mathcal{H}\mathcal{H}^*}$ is a Hermitian matrix, we have

$$H_{\mathcal{H}\mathcal{H}^*} \succeq \mathbf{O} \quad \Leftrightarrow \quad H_{\mathcal{R}\mathcal{R}} \succeq \mathbf{O}, \quad (44)$$

which concludes that (a) equals (b).

Next, to show that (b) is equivalent to (c), $\forall x_{\mathcal{H}} \in \mathcal{H}$, $x_{\mathcal{H}} \neq \mathbf{0}$, we have

$$\begin{aligned} x_{\mathcal{H}}^H H_{\mathcal{H}\mathcal{H}^*} x_{\mathcal{H}} &= \begin{pmatrix} x^i \\ x^j \\ x^k \end{pmatrix}^H \begin{pmatrix} H_{qq^*} & H_{q^i q^*} & H_{q^j q^*} & H_{q^k q^*} \\ H_{qq^{i*}} & H_{q^i q^{i*}} & H_{q^j q^{i*}} & H_{q^k q^{i*}} \\ H_{qq^{j*}} & H_{q^i q^{j*}} & H_{q^j q^{j*}} & H_{q^k q^{j*}} \\ H_{qq^{k*}} & H_{q^i q^{k*}} & H_{q^j q^{k*}} & H_{q^k q^{k*}} \end{pmatrix} \begin{pmatrix} x^i \\ x^j \\ x^k \end{pmatrix} \\ &= \sum_{\mu, \nu \in \{1, i, j, k\}} x^{\mu H} H_{q^{\nu} q^{\mu*}} x^{\nu} \\ &\stackrel{(7)}{=} 4 \sum_{\nu \in \{1, i, j, k\}} \text{Re} \{ x^H H_{q^{\nu} q^*} x^{\nu} \}. \end{aligned} \quad (45)$$

Since $H_{\mathcal{H}\mathcal{H}^*}$ is a Hermitian matrix, we obtain

$$\begin{aligned} \sum_{\nu \in \{1, i, j, k\}} \text{Re} \{ x^H H_{q^{\nu} q^*} x^{\nu} \} &\geq 0, \quad \forall x \in \mathbb{H}^n, x \neq \mathbf{0} \\ &\Leftrightarrow \quad H_{\mathcal{H}\mathcal{H}^*} \succeq \mathbf{O}. \end{aligned} \quad (46)$$

This concludes that (b) equals (c). $\square$

From Theorem 3.5, we obtain the following result.

Corollary 3.1: Consider a convex set $\mathcal{C} \subset \mathbb{H}^n$ and a second-order continuous real-differentiable quaternion function $f(q) : \mathcal{C} \rightarrow \mathbb{R}$. If

$$H_{qq^*} \succeq \mathbf{O}, \text{ and } H_{q^{\nu} q^*} = \mathbf{O}, \quad \nu \in \{i, j, k\}, \quad (47)$$

then $f(q)$ is convex.

Proof: Since $H_{qq^*} \succeq \mathbf{O}$, and $H_{q^{\nu} q^*} = \mathbf{O}$, $\nu \in \{i, j, k\}$, we have $\forall x \in \mathbb{H}^n$, $x \neq \mathbf{0}$,

$$\sum_{\nu \in \{1, i, j, k\}} \text{Re} \{ x^H H_{q^{\nu} q^*} x^{\nu} \} = \text{Re} \{ x^H H_{qq^*} x \} \geq 0. \quad (48)$$

By Theorem 3.5, $f(q)$ is convex. $\square$

Lemma 3.1: The quaternion matrix $A \in \mathbb{H}^{n \times n}$ is positive definite (positive semi-definite), iff all principal submatrices of A are positive definite (positive semi-definite).

Proof: The proof follows the same steps as its counterpart in the real field in Section 7.1 of [40]. $\square$

Upon applying Lemma 3.1, we can obtain a necessary condition for convex quaternion functions.

TABLE I
SEVERAL DERIVATIVES PERFORMED BY THE GHR CALCULUS
FROM TABLE IV OF [26], $\forall A \in \mathbb{H}^{n \times n}$, $\forall a \in \mathbb{H}^n$, $\forall b \in \mathbb{H}^n$,
 $\forall \alpha \in \mathbb{H}$, $\forall \beta \in \mathbb{H}$

$f(q)$ or $f(q)$	$\frac{\partial f}{\partial q}$ or $\frac{\partial f}{\partial q}$	$\frac{\partial f}{\partial q^*}$ or $\frac{\partial f}{\partial q^*}$
$a^T q \beta$	$a^T \text{Re}\{\beta\}$	$-\frac{1}{2} a^T \beta^*$
$\alpha q^H b$	$-\frac{1}{2} \alpha b^H$	$\alpha \text{Re}\{b^T\}$
$A q \beta$	$A \text{Re}\{\beta\}$	$-\frac{1}{2} A \beta^*$
$q^H A q$	$q^H A - \frac{1}{2} (A q)^H$	$-\frac{1}{2} q^H A + \text{Re}\{(A q)^T\}$

Theorem 3.6: Consider a convex set $\mathcal{C} \subset \mathbb{H}^n$ and a second-order continuous real-differentiable quaternion function $f(q) : \mathcal{C} \rightarrow \mathbb{R}$. If $f(q)$ is convex, then

$$H_{qq^*} \succeq \mathbf{O}. \quad (49)$$

Proof: Upon applying Theorem 3.5, together with the convexity of $f(q)$, we have $H_{\mathcal{H}\mathcal{H}^*} \succeq \mathbf{O}$. By (25) and Lemma 3.1, we finally obtain $H_{qq^*} \succeq \mathbf{O}$. $\square$

D. An Example of Convex Quaternion Function

We next provide a basic example to determine the convexity of a quaternion function. In this example, we use certain GHR derivatives presented in Table IV of [26], which are included in Table I here.

Example 3.1: If $f(q) = \|Aq - b\|_2^2$, $\forall q \in \mathbb{H}^n$, $A \in \mathbb{H}^{m \times n}$, $b \in \mathbb{H}^m$, then $f(q)$ is convex.

Proof: (**First-order characterization criterion**) By the definition of 2-norm, we have

$$\begin{aligned} f(q) &= \|Aq - b\|_2^2 \\ &= (Aq - b)^H (Aq - b) \\ &= q^H A^H A q - q^H A^H b - b^H A q + b^H b. \end{aligned} \quad (50)$$

Upon using the first, the second, and the fourth rows of Table I, we take the gradient of $f(q)$ with respect to q^* to yield

$$\begin{aligned} \nabla_{q^*} f(q) &\triangleq \left(\frac{\partial f}{\partial q^*} \right)^T \stackrel{(22)}{=} \left(\frac{\partial f}{\partial q} \right)^H \\ &= \frac{1}{2} A^H A q + \frac{1}{2} A^H b - A^H b \\ &= \frac{1}{2} A^H (A q - b). \end{aligned} \quad (51)$$

Next, $\forall p, q \in \mathbb{H}^n$, we obtain

$$\begin{aligned} f(q) - f(p) - 4 \text{Re} \left\{ \nabla_{p^*} f(p)^H (q - p) \right\} \\ &= (Aq - b)^H (Aq - b) - (Ap - b)^H (Ap - b) \\ &\quad - 2 \text{Re} \left\{ \left(A^H (Ap - b) \right)^H (q - p) \right\} \\ &= q^H A^H A q + p^H A^H A p - p^H A^H A q - q^H A^H A p \\ &= (q - p)^H A^H A (q - p) \\ &= \|A(q - p)\|_2^2 \\ &\geq 0. \end{aligned} \quad (52)$$

Then, by (1), the quaternion function $f(q)$ is convex.

(Gradient monotonicity criterion) $\forall p, q \in \mathbb{H}^n$, we have

$$\begin{aligned} & \operatorname{Re} \left\{ (\nabla_{p^*} f(p) - \nabla_{q^*} f(q))^H (p - q) \right\} \\ &= \frac{1}{2} \operatorname{Re} \left\{ \left(A^H (Ap - b) - A^H (Aq - b) \right)^H (p - q) \right\} \\ &= \frac{1}{2} \operatorname{Re} \left\{ (p - q)^H A^H A (p - q) \right\} \\ &= \frac{1}{2} (p - q)^H A^H A (p - q) \\ &= \frac{1}{2} \|A(q - p)\|_2^2 \\ &\geq 0. \end{aligned} \quad (53)$$

From Theorem 3.1, we know that $f(q)$ is convex.

(Second-order characterization criterion) Using the third row of Table I, we have

$$\begin{aligned} H_{qq^*} &\triangleq \frac{\partial}{\partial q} \left(\frac{\partial f}{\partial q^*} \right)^T = \frac{\partial \nabla_{q^*} f(q)}{\partial q} \\ &\stackrel{(51)}{=} \frac{\partial \left(\frac{1}{2} A^H (Aq - b) \right)}{\partial q} = \frac{1}{2} A^H A \succeq O, \end{aligned} \quad (54)$$

and for any $\nu \in \{i, j, k\}$,

$$\begin{aligned} H_{q^\nu q^*} &\triangleq \frac{\partial}{\partial q^\nu} \left(\frac{\partial f}{\partial q^*} \right)^T = \frac{\partial \nabla_{q^*} f(q)}{\partial q^\nu} \\ &\stackrel{(51)}{=} \frac{\partial \left(\frac{1}{2} A^H (Aq - b) \right)}{\partial q^\nu} \stackrel{(19)}{=} \frac{1}{2} A^H A \frac{\partial q}{\partial q^\nu} = O. \end{aligned} \quad (55)$$

By Corollary 3.1, we know that $f(q)$ is convex. $\square$

IV. DISCRIMINANT THEOREMS FOR STRONGLY CONVEX QUATERNION FUNCTIONS

We shall now discuss the discriminant criteria for strongly convex quaternion functions, building upon the theorems for convexity. These criteria will be helpful for designing optimization algorithms.

Definition 4.1: (Strongly convex function). The quaternion function $f(q) : \mathcal{C} \subset \mathbb{H}^n \rightarrow \mathbb{R}$ is called strongly convex, if $\forall p, q \in \mathcal{C}$, $\forall \theta \in [0, 1]$, $\exists \sigma > 0$,

$$f(\theta p + (1 - \theta)q) \leq \theta f(p) + (1 - \theta)f(q) - \frac{\sigma}{2} \theta(1 - \theta) \|p - q\|_2^2, \quad (56)$$

where σ is the strongly convex parameter. For convenience, $f(q)$ is also called σ -strongly convex.

We obtain the following equivalence theorem based on the definition of strongly convex functions.

Theorem 4.1: The quaternion function $f(q) : \mathcal{C} \subset \mathbb{H}^n \rightarrow \mathbb{R}$ is σ -strongly convex, iff $\exists \sigma > 0$, s.t. the function

$$g(q) \triangleq f(q) - \frac{\sigma}{2} \|q\|_2^2 \quad (57)$$

is convex.

Similar to convex quaternion functions, strongly convex quaternion functions also have first-order characterization, gradient monotonicity, and second-order characterization.

Theorem 4.2: (First-order characterization). Consider a convex set $\mathcal{C} \subset \mathbb{H}^n$ and a real-differentiable quaternion function $f(q) : \mathcal{C} \rightarrow \mathbb{R}$. Then, $f(q)$ is σ -strongly convex iff $\forall p, q \in \mathcal{C}$,

$$f(q) \geq f(p) + 4 \operatorname{Re} \left\{ \nabla_{p^*} f(p)^H (q - p) \right\} + \frac{\sigma}{2} \|q - p\|_2^2. \quad (58)$$

Proof: From Theorem 4.1, $f(p)$ is strongly convex iff $g(p) = f(p) - \frac{1}{2} \sigma \|p\|_2^2$ is convex. By (1), $\forall p, q \in \mathcal{C}$,

$$g(q) \geq g(p) + 4 \operatorname{Re} \left\{ \nabla_{p^*} g(p)^H (q - p) \right\}. \quad (59)$$

Using the fourth row of TABLE I, $\nabla_{p^*} g(p) = \nabla_{p^*} f(p) - \frac{1}{4} \sigma p$. (59) becomes

$$\begin{aligned} & f(q) - \frac{\sigma}{2} \|q\|_2^2 \\ & \geq f(p) - \frac{\sigma}{2} \|p\|_2^2 + 4 \operatorname{Re} \left\{ \left(\nabla_{p^*} f(p) - \frac{\sigma}{4} p \right)^H (q - p) \right\}. \end{aligned} \quad (60)$$

Upon substituting

$$\begin{aligned} & \frac{\sigma}{2} \|q\|_2^2 - \frac{\sigma}{2} \|p\|_2^2 + 4 \operatorname{Re} \left\{ \left(\nabla_{p^*} f(p) - \frac{\sigma}{4} p \right)^H (q - p) \right\} \\ &= 4 \operatorname{Re} \left\{ \nabla_{p^*} f(p)^H (q - p) \right\} - \sigma \operatorname{Re} \{ p^H q \} + \sigma \|p\|_2^2 \\ & \quad + \frac{\sigma}{2} \|q\|_2^2 - \frac{\sigma}{2} \|p\|_2^2 \\ & \stackrel{(12)}{=} 4 \operatorname{Re} \left\{ \nabla_{p^*} f(p)^H (q - p) \right\} + \frac{\sigma}{2} \|q - p\|_2^2 \end{aligned} \quad (61)$$

into (60), we obtain (58). $\square$

Theorem 4.3: (Gradient monotonicity). Consider a convex set $\mathcal{C} \subset \mathbb{H}^n$ and a real-differentiable quaternion function $f(q) : \mathcal{C} \rightarrow \mathbb{R}$. Then, $f(q)$ is σ -strongly convex iff $\forall p, q \in \mathcal{C}$,

$$\operatorname{Re} \left\{ (\nabla_{p^*} f(p) - \nabla_{q^*} f(q))^H (p - q) \right\} \geq \frac{\sigma}{4} \|p - q\|_2^2. \quad (62)$$

Proof: From Theorem 4.1, $f(q)$ is strongly convex iff $g(q) = f(q) - \frac{1}{2} \sigma \|q\|_2^2$ is convex. Upon applying Theorem 3.1, we have

$$\operatorname{Re} \left\{ (\nabla_{p^*} g(p) - \nabla_{q^*} g(q))^H (p - q) \right\} \geq 0, \quad \forall p, q \in \mathcal{C}. \quad (63)$$

Using the fourth row of TABLE I, we have $\nabla_{q^*} g(q) = \nabla_{q^*} f(q) - \frac{1}{4} \sigma q$. Then (63) becomes

$$\operatorname{Re} \left\{ \left(\nabla_{p^*} f(p) - \frac{\sigma}{4} p - \nabla_{q^*} f(q) + \frac{\sigma}{4} q \right)^H (p - q) \right\} \geq 0. \quad (64)$$

After rearranging the terms in (64), we obtain (62). $\square$

Theorem 4.4: (Second-order characterization). Consider a convex set $\mathcal{C} \subset \mathbb{H}^n$ and a second-order continuous real-differentiable quaternion function $f(q) : \mathcal{C} \rightarrow \mathbb{R}$. Then, the following three statements are equivalent:

- (a) $f(q)$ is σ -strongly convex;
- (b) $H_{\mathcal{H}\mathcal{H}^*} \succeq \frac{1}{4} \sigma I_{4n}$;
- (c) $\sum_{\nu \in \{1, i, j, k\}} \operatorname{Re} \{ x^H H_{q^\nu q^*} x^\nu \} - \frac{1}{4} \sigma \|x\|_2^2 \geq 0, \quad \forall x \in \mathbb{H}^n, x \neq 0$.

Proof: We first show that (a) is equivalent to (b). Define $g(\mathbf{q}) \triangleq f(\mathbf{q}) - \frac{1}{2}\sigma\|\mathbf{q}\|_2^2$, $h(\mathbf{q}) \triangleq 2\|\mathbf{q}\|_2^2 = 2\mathbf{q}^H\mathbf{q} = 2(\mathbf{q}^H\mathbf{q})^\mu \stackrel{(4)}{=} 2\mathbf{q}^{\mu H}\mathbf{q}^\mu$, $\mu \in \{1, i, j, k\}$. From the fourth row of TABLE I, we have

$$\left(\frac{\partial h}{\partial \mathbf{q}^{\mu*}}\right)^\top = \mathbf{q}^\mu, \quad \mu \in \{1, i, j, k\}. \quad (65)$$

Then, $\forall \mu \in \{1, i, j, k\}$,

$$\frac{\partial}{\partial \mathbf{q}^\mu} \left(\frac{\partial h}{\partial \mathbf{q}^{\mu*}}\right)^\top \stackrel{(65)}{=} \frac{\partial \mathbf{q}^\mu}{\partial \mathbf{q}^\mu} = \mathbf{I}_n, \quad (66)$$

and $\forall \mu, \nu \in \{1, i, j, k\}$, $\mu \neq \nu$,

$$\frac{\partial}{\partial \mathbf{q}^\nu} \left(\frac{\partial h}{\partial \mathbf{q}^{\mu*}}\right)^\top \stackrel{(65)}{=} \frac{\partial \mathbf{q}^\mu}{\partial \mathbf{q}^\nu} = \mathbf{O}. \quad (67)$$

By (25), the augmented quaternion Hessian matrix of h is $\mathbf{I}_{4n}$. Therefore, the augmented quaternion Hessian matrix of g is $\mathbf{H}_{\mathcal{H}\mathcal{H}^*} - \frac{1}{4}\sigma\mathbf{I}_{4n}$. By Theorem 3.5, we conclude that (a) is equivalent to (b).

Next, to show that (b) is equivalent to (c), $\forall \mathbf{x}_{\mathcal{H}} \in \mathcal{H}$, $\mathbf{x}_{\mathcal{H}} \neq \mathbf{0}$, we have

$$\begin{aligned} & \mathbf{x}_{\mathcal{H}}^H \left(\mathbf{H}_{\mathcal{H}\mathcal{H}^*} - \frac{\sigma}{4}\mathbf{I}_{4n} \right) \mathbf{x}_{\mathcal{H}} \\ &= \mathbf{x}_{\mathcal{H}}^H \mathbf{H}_{\mathcal{H}\mathcal{H}^*} \mathbf{x}_{\mathcal{H}} - \frac{\sigma}{4} \mathbf{x}_{\mathcal{H}}^H \mathbf{x}_{\mathcal{H}} \\ &\stackrel{(11)}{=} \sum_{\mu, \nu \in \{1, i, j, k\}} \mathbf{x}^{\mu H} \mathbf{H}_{\mathbf{q}^\nu \mathbf{q}^{\mu*}} \mathbf{x}^\nu - \sigma \mathbf{x}^H \mathbf{x} \\ &\stackrel{(7)}{=} 4 \sum_{\nu \in \{1, i, j, k\}} \operatorname{Re} \{ \mathbf{x}^H \mathbf{H}_{\mathbf{q}^\nu \mathbf{q}^*} \mathbf{x}^\nu \} - \sigma \|\mathbf{x}\|_2^2. \end{aligned} \quad (68)$$

Since $\mathbf{H}_{\mathcal{H}\mathcal{H}^*}$ is a Hermitian matrix, $\mathbf{H}_{\mathcal{H}\mathcal{H}^*} - \frac{1}{4}\sigma\mathbf{I}_{4n}$ is also a Hermitian matrix. $\forall \mathbf{x} \in \mathbb{H}^n$, $\mathbf{x} \neq \mathbf{0}$,

$$\begin{aligned} & \sum_{\nu \in \{1, i, j, k\}} \operatorname{Re} \{ \mathbf{x}^H \mathbf{H}_{\mathbf{q}^\nu \mathbf{q}^*} \mathbf{x}^\nu \} - \frac{\sigma}{4} \|\mathbf{x}\|_2^2 \geq 0, \\ & \Leftrightarrow \mathbf{H}_{\mathcal{H}\mathcal{H}^*} - \frac{\sigma}{4}\mathbf{I}_{4n} \succeq \mathbf{O}. \end{aligned} \quad (69)$$

This concludes that (b) is equivalent to (c). $\square$

From Theorem 4.4, we obtain the following result.

Corollary 4.1: Consider a convex set $\mathcal{C} \subset \mathbb{H}^n$ and a second-order continuous real-differentiable quaternion function $f(\mathbf{q}) : \mathcal{C} \rightarrow \mathbb{R}$. If

$$\mathbf{H}_{\mathbf{q}\mathbf{q}^*} \succeq \frac{\sigma}{4}\mathbf{I}_n, \quad \text{and } \mathbf{H}_{\mathbf{q}^\nu \mathbf{q}^*} = \mathbf{O}, \quad \nu \in \{i, j, k\}, \quad (70)$$

then $f(\mathbf{q})$ is σ -strongly convex.

By Theorem 4.4 and Lemma 3.1, we can obtain a necessary condition for σ -strongly convex quaternion functions.

Theorem 4.5: Consider a convex set $\mathcal{C} \subset \mathbb{H}^n$ and a second-order continuous real-differentiable quaternion function $f(\mathbf{q}) : \mathcal{C} \rightarrow \mathbb{R}$. If $f(\mathbf{q})$ is σ -strongly convex, then

$$\mathbf{H}_{\mathbf{q}\mathbf{q}^*} \succeq \frac{\sigma}{4}\mathbf{I}_n. \quad (71)$$

Proof: Note that $f(\mathbf{q})$ is σ -strongly convex, and upon applying Theorem 4.4, we have $\mathbf{H}_{\mathcal{H}\mathcal{H}^*} \succeq \frac{1}{4}\sigma\mathbf{I}_{4n}$. By (25) and Lemma 3.1, we finally obtain $\mathbf{H}_{\mathbf{q}\mathbf{q}^*} \succeq \frac{1}{4}\sigma\mathbf{I}_n$. $\square$

Theorem 4.6: (Quadratic termination). For the positive definite quadratic function $f(\mathbf{q}) = \mathbf{q}^H \mathbf{A} \mathbf{q} + \operatorname{Re}\{\mathbf{b}^H \mathbf{q}\} + c$, $\mathbf{b} \in \mathbb{H}^n$, $c \in \mathbb{R}$, and the matrix $\mathbf{A} \in \mathbb{H}^{n \times n}$ is Hermitian and positive definite. Then, the quaternion Newton (QN) method converges with one step.

Proof: From (51), (54), and (55), it is straightforward to compute the gradient of $f(\mathbf{q})$

$$\nabla_{\mathbf{q}^*} f(\mathbf{q}) = \frac{1}{2} \mathbf{A} \mathbf{q} + \frac{1}{4} \mathbf{b}, \quad (72)$$

and the Hessian matrices of $f(\mathbf{q})$

$$\mathbf{H}_{\mathbf{q}\mathbf{q}^*} = \frac{\partial \nabla_{\mathbf{q}^*} f(\mathbf{q})}{\partial \mathbf{q}} = \frac{1}{2} \mathbf{A}, \quad (73)$$

for any $\nu \in \{i, j, k\}$,

$$\mathbf{H}_{\mathbf{q}^\nu \mathbf{q}^*} = \frac{\partial \nabla_{\mathbf{q}^*} f(\mathbf{q})}{\partial \mathbf{q}^\nu} = \mathbf{O}. \quad (74)$$

By setting $\nabla_{\mathbf{q}^*} f(\mathbf{q}) = \mathbf{0}$ as per (72), we can obtain the closed-form solution

$$\mathbf{q}_{opt} = -\frac{1}{2} \mathbf{A}^{-1} \mathbf{b}. \quad (75)$$

By (74), the update rule of QN [25] is expressed as

$$\begin{aligned} \mathbf{q}^1 &= \mathbf{q}^0 - \mathbf{H}_{\mathbf{q}\mathbf{q}^*}^{-1} \nabla_{\mathbf{q}^*} f(\mathbf{q}^0) \\ &= \mathbf{q}^0 - \mathbf{A}^{-1} \left(\mathbf{A} \mathbf{q}^0 + \frac{1}{2} \mathbf{b} \right) = -\frac{1}{2} \mathbf{A}^{-1} \mathbf{b} \\ &\stackrel{(75)}{=} \mathbf{q}_{opt}, \end{aligned} \quad (76)$$

where $\mathbf{q}^0$ is initial vector. Therefore, the QN method reaches the optimal solution within one iteration for the positive definite quadratic function, i.e., quadratic termination. $\square$

V. APPLICATIONS OF CONVEX QUATERNION OPTIMIZATION IN SIGNAL PROCESSING

This section analyzes and solves two representative applications using the proposed convex quaternion optimization theory in this paper.

Application 5.1: (QCRC). The quaternion collaborative representation-based classification (QCRC) effectively captures the structural correlation among color channels in color images [41], [42]. The QCRC model can be expressed as

$$\min_{\mathbf{q} \in \mathbb{H}^n} f(\mathbf{q}) = \|\mathbf{A} \mathbf{q} - \mathbf{b}\|_2^2 + \lambda \|\mathbf{q}\|_2^2, \quad (77)$$

where $\mathbf{A} \in \mathbb{H}^{m \times n}$ is a matrix consisting of input features, $\mathbf{b} \in \mathbb{H}^m$ is a vector comprising labels, and $\lambda > 0$. By (51), (54), and (55), the gradient of $f(\mathbf{q})$ is

$$\nabla_{\mathbf{q}^*} f(\mathbf{q}) = \frac{1}{2} \mathbf{A}^H (\mathbf{A} \mathbf{q} - \mathbf{b}) + \frac{1}{2} \lambda \mathbf{q}, \quad (78)$$

and the Hessian matrices of $f(\mathbf{q})$ are

$$\mathbf{H}_{\mathbf{q}\mathbf{q}^*} = \frac{\partial \nabla_{\mathbf{q}^*} f(\mathbf{q})}{\partial \mathbf{q}} = \frac{1}{2} \mathbf{A}^H \mathbf{A} + \frac{1}{2} \lambda \mathbf{I}_n, \quad (79)$$

and

$$\mathbf{H}_{\mathbf{q}^\nu \mathbf{q}^*} = \frac{\partial \nabla_{\mathbf{q}^*} f(\mathbf{q})}{\partial \mathbf{q}^\nu} = \mathbf{O}, \quad \nu \in \{i, j, k\}. \quad (80)$$

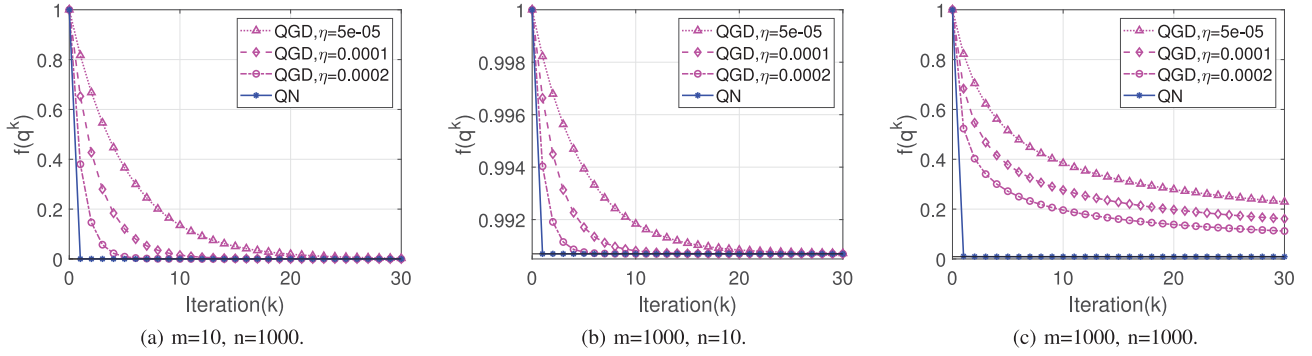


Fig. 2. Numerical validation of quaternion gradient descent (QGD) algorithm and quaternion Newton (QN) method for QCRC, over a range of parameters. Observe that the QN method (blue line) converges to $f(\mathbf{q}_{opt})$ (solid black line) in one step.

Since $\mathbf{A}^H \mathbf{A} \succeq \mathbf{O}$, we have $\mathbf{H}_{\mathbf{q}\mathbf{q}^*} \succeq \frac{1}{2} \lambda \mathbf{I}_n$. According to Corollary 4.1, $f(\mathbf{q})$ is 2λ -strongly convex. By setting $\nabla_{\mathbf{q}^*} f(\mathbf{q}) = \mathbf{0}$ as per (78), we obtain the closed-form solution

$$\mathbf{q}_{opt} = (\mathbf{A}^H \mathbf{A} + \lambda \mathbf{I}_n)^{-1} \mathbf{A}^H \mathbf{b}. \quad (81)$$

Next, we aim to test the convergence performance of the quaternion iteration algorithms for solving the QCRC problem. In particular, the QN method can converge in one step, so its computational cost is equivalent to (81). However, many optimization problems are too complex to have closed-form solutions, so iterative algorithms will have to be used. By the strong convexity of $f(\mathbf{q})$, we can employ the quaternion gradient descent (QGD) algorithm and the QN method [25] to find the optimal solution to problem (77). The update rule of QGD is

$$\begin{aligned} \mathbf{q}^{t+1} &= \mathbf{q}^t - 4\eta \nabla_{\mathbf{q}^*} f(\mathbf{q}^t) \\ &= \mathbf{q}^t - 2\eta (\mathbf{A}^H (\mathbf{A} \mathbf{q}^t - \mathbf{b}) + \lambda \mathbf{q}^t), \end{aligned} \quad (82)$$

where $\eta > 0$ is the step size. By (80), the update rule of QN can be expressed as

$$\begin{aligned} \mathbf{q}^{t+1} &= \mathbf{q}^t - \mathbf{H}_{\mathbf{q}\mathbf{q}^*}^{-1} \nabla_{\mathbf{q}^*} f(\mathbf{q}^t) \\ &= \mathbf{q}^t - (\mathbf{A}^H \mathbf{A} + \lambda \mathbf{I}_n)^{-1} (\mathbf{A}^H (\mathbf{A} \mathbf{q}^t - \mathbf{b}) + \lambda \mathbf{q}^t). \end{aligned} \quad (83)$$

All algorithms employed the same data matrix $\mathbf{A}$, vector $\mathbf{b}$, initial vector $\mathbf{q}^0 = \mathbf{0}$, and $2\lambda = 0.1$, where λ can be optimized by grid search [42], and we follow 3D Basis Pursuit Denoising [22] to set $\lambda = 0.05$. We generated a matrix $\mathbf{A} \in \mathbb{H}^{10 \times 1000}$ and a vector $\mathbf{b} \in \mathbb{H}^{10}$, where the four real parts of $\mathbf{A}$ are i.i.d. sampled from a unit Gaussian distribution and $\mathbf{b}$ is normalized with 2-norm, in Fig. 2(a). In Fig. 2(b) and (c), $\mathbf{A}$ and $\mathbf{b}$ were generated in the same way, but with different scales: $\mathbf{A} \in \mathbb{H}^{1000 \times 10}$ and $\mathbf{b} \in \mathbb{H}^{1000}$ are in (b); $\mathbf{A} \in \mathbb{H}^{1000 \times 1000}$ and $\mathbf{b} \in \mathbb{H}^{1000}$ are in (c). The numerical experiments of this application and subsequent experiments were completed using the Quaternion Toolbox for Matlab¹ (QTFM).

The performance curves of the objective function for 30 iterations are plotted in Fig. 2. We can clearly see that $f(\mathbf{q}^k)$ converges to $f(\mathbf{q}_{opt})$, but $f(\mathbf{q}_{opt}) \neq 0$ in Fig. 2(b) because it is underdetermined. Larger step sizes result in a faster descent

in the QGD algorithm, but the performance of the QN method is the best. Compared with the closed-form solution, the QGD algorithm has lower computational cost per step because there is no matrix inversion operation. The QN method converges in one step, consistent with Theorem 4.6, so its calculation cost is the same as the closed-form solution. QN can also solve optimization problems without closed-form solutions.

Application 5.2: (QBLR). Logistic regression is the most fundamental linear classification model. We considered a quaternion binary logistic regression (QBLR) model

$$\min_{\mathbf{q} \in \mathbb{H}^m} \ell(\mathbf{q}) = \frac{1}{n} \sum_{i=1}^n \ln(1 + \exp(-b_i \operatorname{Re}\{\mathbf{a}_i^H \mathbf{q}\})) + \lambda \|\mathbf{q}\|_2^2. \quad (84)$$

where the provided data pairs $\{\mathbf{a}_i, b_i\}_{i=1}^n$ are independent and identically distributed, $\mathbf{a}_i \in \mathbb{H}^m$, $b_i \in \{1, -1\}$, and $\lambda > 0$. It can be showed that

$$\begin{aligned} \nabla_{\mathbf{q}^*} \ell(\mathbf{q}) &= \frac{1}{2n} \sum_{i=1}^n \frac{\exp(-b_i \operatorname{Re}\{\mathbf{a}_i^H \mathbf{q}\})}{1 + \exp(-b_i \operatorname{Re}\{\mathbf{a}_i^H \mathbf{q}\})} (-b_i \mathbf{a}_i) + \frac{1}{2} \lambda \mathbf{q} \\ &= -\frac{1}{2n} \sum_{i=1}^n (1 - g_i(\mathbf{q})) b_i \mathbf{a}_i + \frac{1}{2} \lambda \mathbf{q}, \end{aligned} \quad (85)$$

where $g_i(\mathbf{q}) = \frac{1}{1 + \exp(-b_i \operatorname{Re}\{\mathbf{a}_i^H \mathbf{q}\})}$. Further, the Hessian matrices of $\ell(\mathbf{q})$ are

$$\begin{aligned} \mathbf{H}_{\mathbf{q}\mathbf{q}^*} &= \frac{1}{2n} \sum_{i=1}^n b_i \mathbf{a}_i \frac{\partial g_i(\mathbf{q})}{\partial \mathbf{q}} + \frac{1}{2} \lambda \mathbf{I}_m \\ &= \frac{1}{4n} \sum_{i=1}^n b_i \frac{\exp(-b_i \operatorname{Re}\{\mathbf{a}_i^H \mathbf{q}\})}{(1 + \exp(-b_i \operatorname{Re}\{\mathbf{a}_i^H \mathbf{q}\}))^2} b_i \mathbf{a}_i \mathbf{a}_i^H + \frac{\lambda}{2} \mathbf{I}_m \\ &= \frac{1}{4n} \sum_{i=1}^n (1 - g_i(\mathbf{q})) g_i(\mathbf{q}) \mathbf{a}_i \mathbf{a}_i^H + \frac{\lambda}{2} \mathbf{I}_m, \end{aligned} \quad (86)$$

and for any $\nu \in \{i, j, k\}$,

$$\mathbf{H}_{\mathbf{q}^\nu \mathbf{q}^*} = \mathbf{O}. \quad (87)$$

Since $\mathbf{a}_i \mathbf{a}_i^H \succeq \mathbf{O}$ in (86), we know $\mathbf{H}_{\mathbf{q}\mathbf{q}^*} \succeq \frac{1}{2} \lambda \mathbf{I}_m$. By Corollary 4.1, we conclude that $\ell(\mathbf{q})$ is 2λ -strongly convex.

Using the matrix notation, we can express the above results in a more compact form. By defining the matrix

¹QTFM: <https://sourceforge.net/projects/qtfm/>

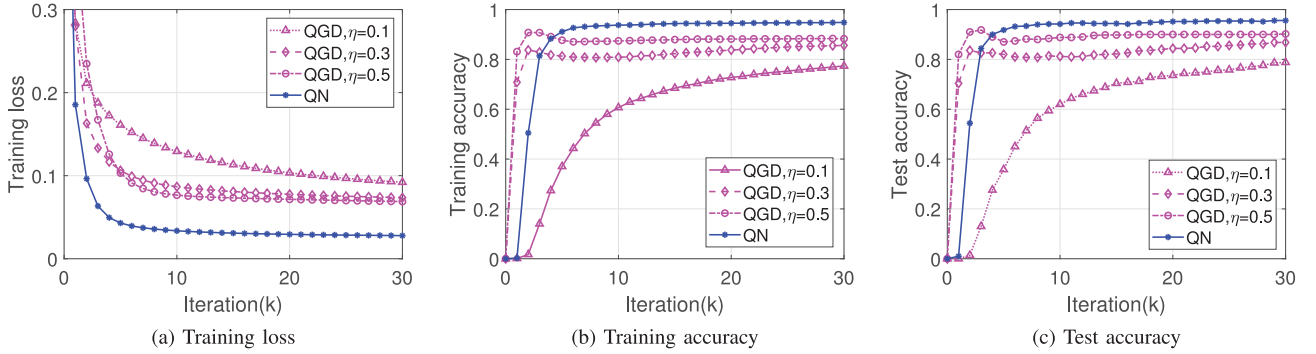


Fig. 3. Numerical validation of QGD and QN for QBLR.

$\mathbf{A} = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n) \in \mathbb{H}^{m \times n}$, the vector $\mathbf{b} = (b_1, b_2, \dots, b_n)^T$, and $\mathbf{g}(\mathbf{q}) = (g_1(\mathbf{q}), g_2(\mathbf{q}), \dots, g_n(\mathbf{q}))^T$, the gradient and the Hessian matrix of $\ell(\mathbf{q})$ can be rewritten as

$$\nabla_{\mathbf{q}^*} \ell(\mathbf{q}) = -\frac{1}{4n} \mathbf{A}(\mathbf{b} - \mathbf{b} \odot \mathbf{g}(\mathbf{q})) + \frac{1}{2} \lambda \mathbf{q}, \quad (88)$$

$$H_{\mathbf{q}\mathbf{q}^*} = \frac{1}{4n} \mathbf{A}W(\mathbf{q})\mathbf{A}^H + \frac{1}{2} \lambda \mathbf{I}_m, \quad (89)$$

where $\odot$ denotes the Hadamard product, and $W(\mathbf{q})$ is a diagonal matrix generated by $\{g_i(\mathbf{q})(1 - g_i(\mathbf{q}))\}_{i=1}^n$.

We next solve the optimization problem (84) using the QGD algorithm and the QN method [25]. The update rule of the QGD algorithm is given by

$$\begin{aligned} \mathbf{q}^{t+1} &= \mathbf{q}^t - 4\eta \nabla_{\mathbf{q}^*} \ell(\mathbf{q}^t) \\ &= \mathbf{q}^t + \frac{\eta}{n} \mathbf{A}(\mathbf{b} - \mathbf{b} \odot \mathbf{g}(\mathbf{q}^t)) - 2\eta \lambda \mathbf{q}^t, \end{aligned} \quad (90)$$

where $\eta > 0$ is the step size. From (87), the update rule of the QN method can be expressed as

$$\begin{aligned} \mathbf{q}^{t+1} &= \mathbf{q}^t - H_{\mathbf{q}\mathbf{q}^*}^{-1} \nabla_{\mathbf{q}^*} \ell(\mathbf{q}^t) = \mathbf{q}^t \\ &+ (\mathbf{A}W(\mathbf{q}^t)\mathbf{A}^H + 2\lambda n \mathbf{I}_m)^{-1} (\mathbf{A}(\mathbf{b} - \mathbf{b} \odot \mathbf{g}(\mathbf{q}^t)) - 2\lambda n \mathbf{q}^t). \end{aligned} \quad (91)$$

We selected a dataset of ECG sequence², split the available 5000 data into a training set of 4500 data and a test set of 500 data, and combined 140 real features into 35 quaternion features. Different combinations may slightly affect classification performances, but do not affect the verification of convergence results. The results of numerical experiments for 30 iterations are plotted in Fig. 3 with $2\lambda = 0.1$. Observe that the training loss of QN and QGD gradually decreased as k increased. Since the objective function of QBLR is not quadratic, QN did not converge within one step, but it still exhibited better convergence than QGD. The test accuracy of QN gradually approached 96% as k increased, and QN finally reached the highest test accuracy, as shown in Fig. 3.

VI. CONCLUSION

We have established the theory of convex optimization in quaternion variables. This has been achieved based on the

GHR calculus, an enabling methodology in quaternion signal processing and machine learning. This has resulted in the development of several discriminant theorems for convex functions in the quaternion field, utilizing (7), (23), (24), (25), and (26). Furthermore, we have provided several discriminant criteria for strongly convex functions of quaternions by employing the results for convex quaternion functions. In addition, we have proven that the quaternion Newton method can converge in a single step for positive definite quadratic functions. The analysis has been supported by two signal processing applications using the proposed convex quaternion optimization theories, which have enriched the theory of convex quaternion optimization and established a theoretical foundation for quaternion signal processing. While the convexity of non-differentiable quaternion functions by the GHR calculus remains an open problem, our work has provided a foundation and an avenue for further research in this direction.

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²ECG sequence: <https://www.kaggle.com/datasets/salsabilahmid/ecgtxt>

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